

TO: ME 2171 Instrumentation Lab Instructional Staff
FROM: Mateo Jaime Tecum Jacinto, Section 07- Wednesday
SUBJECT: Lab 6, 3-Axis MEMS Accelerometer

February 24th, 2026

1.0 Statement of Purpose

The purpose of this experiment was to measure the motion of an accelerometer and have accurate depictions from the LabVIEW vi. Using the kinematics equations and the accelerometer as the input, the accelerometer sent input signals to the DAQ board that was then outputted into LabVIEW for analysis.

2.1 Background

The ADXL335 iMEMS(microelectromechanical system)) 3-Axis Accelerometer and most accelerometers are made by having layers of plates that are made of conductive material, which are cut into particular shapes. These shapes are referred to as fingers, and the fingers of the plates are mirrors of the other fingers which means that they can all fit within another without contact. A set of fingers remains stationary for the accelerometer and another set moves a certain amount, and the nonstationary set moves closer or further from the stationary set when it is experiencing a force. This displacement is what is used as the measurand. When the displacement occurs, the conductive material also causes a change of capacitance in the circuit of the accelerometer, which results in a change of voltage, which can be measured and recorded on an A/D converter. For this setup the DAQ board received the voltage signals.

According to the data sheet of the ADXL 335 Accelerometer, the accelerometer can output a range of voltages when powered with 1.8-3.6V. The accelerometer has a range of $\pm 3g$'s, and can withstand up to 10,000 g's. It can record up to a frequency range of 0.5-1600Hz on the X & Y axes, and for the Z axis, it ranges from 0.5-550Hz.

Devices manufactured by this technology (the ADXL335 accelerometer) allowed for more devices to be manufactured with accelerometers in the system that were not expensive to make. The iMEMS are made fairly small, so the devices made such as Wii controllers could keep them in the remotes without it hindering the size of the controller. Another application of the tiny cheap accelerometer in devices involves tablets, as accelerometers allowed for the quality-of-life automatic screen rotation to occur by having the device sense the gravity's direction.

3.0 Discussion

3.1 Equipment

- ADXL335 iMEMS 3-Axis Accelerometer
- wires

- Female to male wires
- Siglent SDG1022X Plus function generator
- Banana cables with alligator clips
- LabVIEW computer
- NI-USB-6221 DAQ Board
- Hook wires

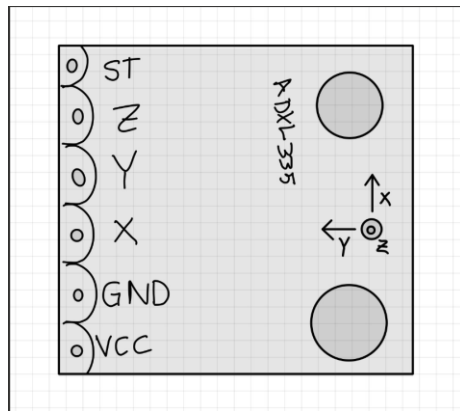


Figure 1. ADXL335 accelerometer drawing

3.2.1

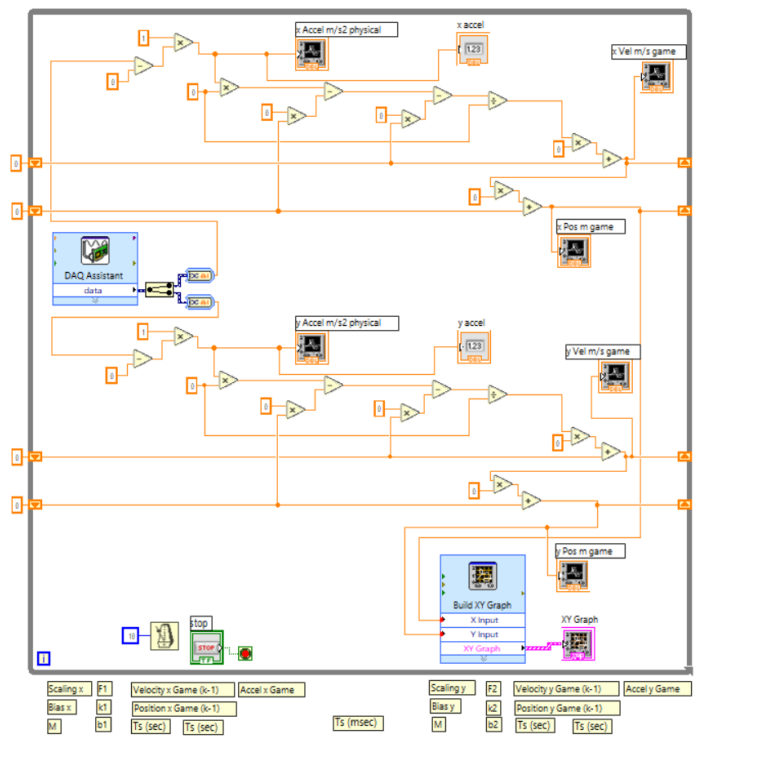


Figure 2. LabVIEW block diagram. Receives voltage data from the accelerometer and using kinetic equations, outputs accel., velocity, force, and position data.

3.2.2

$$F_1(k) = Ma_x(k) \quad (1)$$

$$a_{x,game}(k) = \frac{1}{M}[F_1(k) - b_1v_{x,game}(k-1) - k_1x_{game}(k-1)](2)$$

$$v_{x,game}(k) = v_{x,game}(k-1) + T_s a_{x,game}(k)(3)$$

$$x_{game}(k) = x_{game}(k-1) + T_s v_{x,game}(k)(4)$$

As shown in Eq 1., F_1 is the force that is done to the virtual object in the experiments listed later on in the report, k represents the spring constant, M represents the mass of the virtual object, and a_x represents the acceleration in the x axis. This equation is just Newton's Second Law of Motion, where the force is calculated from the product of the mass of the object and the acceleration.

As shown in Eq 2., the variables are similar to Eq 1., with v_x representing the velocity of the object, $(k-1)$ representing the previous iterations data and $b_1 \cdot v$ representing the velocity and dampening. The dampening coefficient b_1 , limits the motion in the simulation. Eq 2. Is used to calculate the acceleration in the x-dir using another simple kinematics equation and dividing the entire term by the mass of the object.

As shown in Eq 3., the variables are the same as before with the calculation of velocity in the x direction being derived from the previous velocity's iteration ($v(k-1)$) and the sum of the product of the acceleration and the time, T_s . The variable T_s is the time it takes the program to begin measuring data, in units of seconds.

As shown in Eq 4., the displacement in the x direction is calculated with the sum of the previous iteration's displacement and the velocity multiplied by the time, T_s .

It is important to note for $T_s \cdot a$ and $T_s \cdot v$, that both of these represent the integration value of velocity/acceleration, so in Eq 3 and 4, both the dampening coefficient and the spring constant are still being used in these equations implicitly.

3.2.3

The LabVIEW program for this lab has a sampling frequency running at 100Hz. We have this set to 100Hz because we want the program to wait a period of 10ms (inverse of 100Hz). This 10ms period is what we would then use for our calculations of time in our aforementioned equations. Additionally, in this experiment, the bandwidth of the accelerometer, according to the data sheet, is at 50Hz for all axes. Since all axes are at 50Hz, per the Nyquist criterion we want the sampling frequency to be two times the bandwidth frequency to prevent aliasing.

3.3 Calibration

The calibration equations listed below are used in this experiment to calibrate our accelerometer.

$$a_x(k) = K_x[voltage_x(k) - Bias_x] \quad (x1)$$

$$a_y(k) = K_y[voltage_y(k) - Bias_y] \quad (y1)$$

We used them in the beginning of our experiment to find the voltages when the accelerometers x and y directions were acted upon 0g. Our group performed the calibration both at 2.5V and 3.3V in order to observe the ratiometric behavior of the accelerometer. For the calibration equations, the bias values for both is the maximum voltage output that the accelerometer has added to the minimum voltage and divided by two, which is referred to as the the known acceleration value(gravity). The calibration itself involved holding the accelerometer so that the y-direction had a positive value of gravity acting on it, and then again with the x-axis(and also perform two more times with negative gravity on both of these axes). By recording these values, we could calculate the bias and voltage.

For this calibration procedure, at 2.5V, we obtained a value of 1.501V with +g in the x direction. For a -g value at 2.5V in the x axis, we obtained .996V. At 3.3V we obtained a value of 1.33V(+g in x-dir) and 1.96V(-g in x-dir). Using the bias equation (voltage-max + voltage min)/2, we obtained a bias of 1.64V at 3.3V

$$Bias_x = \frac{1.96V + 1.33V}{2} = 1.645V$$

For the y-direction, we also repeated the calibration procedure. At 2.5V, a positive gravity value(+g) resulted in 0.9931V (voltage min), and a negative gravity value (-g) resulted in 1.49405V(voltage min). At 3.3V, +g resulted in 1.312V, and a -g value resulted in 1.996V. Using the 3.3V values, the bias for the y-direction was calculated:

$$Bias_y = \frac{1.996V + 1.312V}{2} = 1.654V$$

To verify our Bias values for both x and y directions, we can look at the range in the data sheet for the bias voltage at 3V, which falls around 1.3-1.65V. For both our x and y direction's bias, they fell around the maximum range of ~1.65V, which makes sense as our

voltage is higher than 3V. Now to obtain our K values for both directions we used our known force value(gravity) and calculated the scale factors(sensitivity) below:

$$K_y = \left| \frac{2(9.81)}{(1.312V - 1.996V)} \right| = 28.68 \frac{m}{s^2 * V}$$

$$K_x = \left| \frac{2(9.81)}{(1.33V - 1.96V)} \right| = 31.142 \frac{m}{s^2 * V}$$

Using the Kx and Ky values, the sensitivity was calculated for both axes below(in units of mV/g):

$$Sensitivity_x = \left| \frac{1.33V - 1.96V}{2} \right| * 1000 = 315 \frac{mV}{g}$$

$$Sensitivity_y = \left| \frac{1.312V - 1.996V}{2} \right| * 1000 = 342 \frac{mV}{g}$$

The sensitivity value range given in the data sheet at 3V ranges from 270-330 mV/g. so with our higher bias at 3.3V, sensitivity ranges that are higher than this range do make sense in this case and are expected, because the accelerometer has ratiometric behavior, which means with more voltage being powered to the accelerometer, the sensitivity and bias will increase in value, which is what we see.

3.4 Exercise 1

In exercise 1, with no spring constants nor dampening coefficients having an effect on the system, as both were set to zero, the only set condition was the mass at 65 grams. From the vi, we saw the virtual ball rolling in the direction that the accelerometer was set to. For example, when the vi began running the accelerometer was set perpendicular to the screen, which means that the ball would not move and just stay still. Once our team tilted the accelerometer in any direction from there, the ball would begin rolling in that direction

3.5 Exercise 2

For exercise 2, we set k=1 and b=0. This led to the same result as exercise 1, where the ball would roll in the direction of the accelerometer's acceleration, but with a spring in play. This spring would pull the ball back to the center, although hard to see. This can be thought of as the Force of the spring opposing the motion of the ball. From the free body diagram below, we can see that the two forces oppose each other.

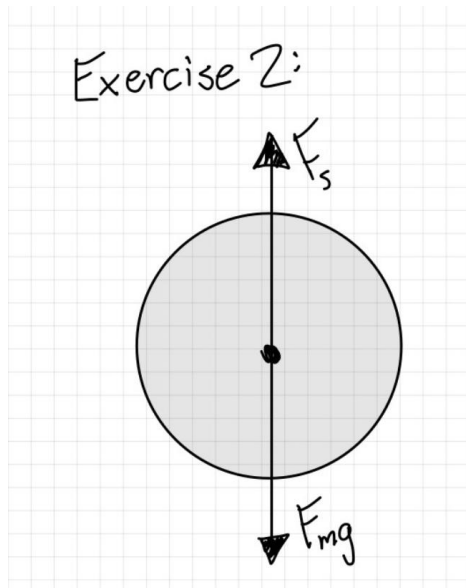


Figure 3. Free Body Diagram of the Force of the spring and Force of gravity acting on the Virtual Mass in Exercise 2

3.6 Exercise 3

In Exercise 3, the spring constant(k) for both the x and y axes were set to 1, and the dampening coefficient (b) was set to 0.1. The procedure involved tilting the accelerometer toward the positive x -direction and attempting to keep it that way. Due to the dampening force and the Spring force, the resulting trajectory of the virtual ball on the graph in the front panel appeared to be a dot oscillating at an end of a spring until it reached an equilibrium. Due to the dampening force opposing the Force of the spring and other motion, the ball would not be in the same height as time progressed. Once equilibrium was reached, the gravity and spring force were equal. The forces are shown in the free body diagram shown below as well as the graph of the ball's motion in figure 5:

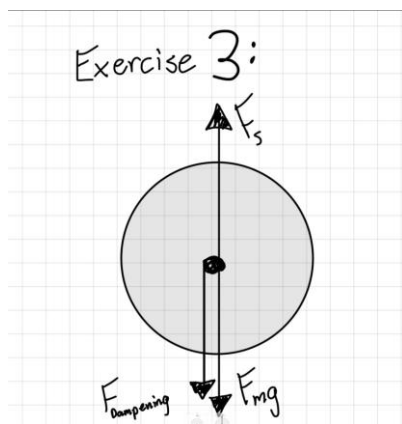


Figure 4. Free body Diagram of the Dampening force, Spring force, and gravity force, in exercise 3

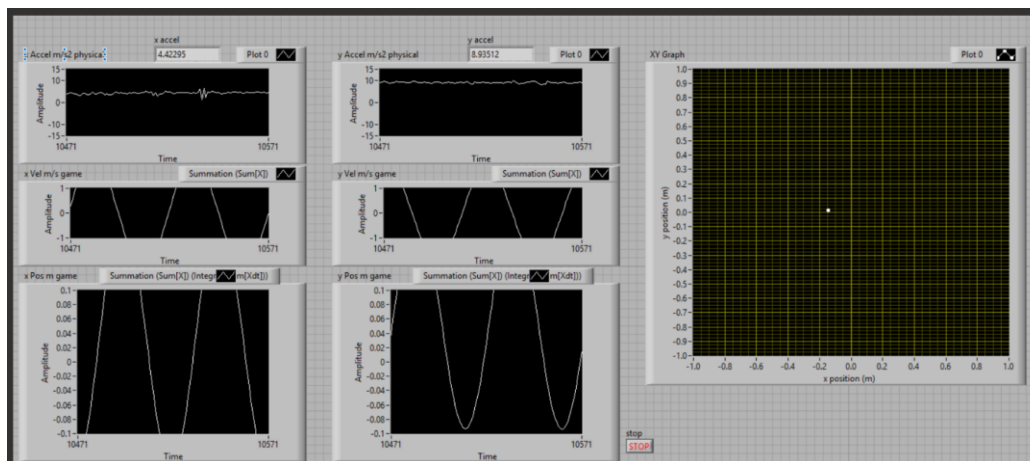


Figure 5. Front panel image of the balls motion in the x-y axes and the displacement on the right

$$\begin{aligned} \sum F(k) &= F_i(k) - b_i \cdot v_{y, \text{game}}(k-1) - k_i y_{\text{game}}(k-1) \\ &\rightarrow \text{At equilibrium } v_y = 0 \\ \sum F(k) &= F_i(k) - k_i y_{\text{game}}(k-1) = 0 \\ F_i(k) &= k_i y_{\text{game}}(k-1) \\ y_{\text{game}} &\rightarrow y_{\text{game}} = \frac{m g}{k_i} = \frac{0.065 \text{ kg} (9.81 \frac{\text{m}}{\text{s}^2})}{1 \text{ N/m}} \\ y &= 0.064 \text{ m} \end{aligned}$$

Figure 6. Exercise 3 calculations

Our group additionally calculated the ball's motion using Equation 2 to find the displacement of the virtual ball. This was done to see if the calculated displacements turned out to be the same value, and looking at the values, the v_i had a larger value which can be explained by any noise that may have been caught by the accelerometer or from integration errors.

3.7 Exercise 4

In this exercise, we had the spring constant set to 1 and the dampening set to 1. This meant we had 3 forces acting on the virtual ball. The procedure also involved tilting the unit towards the x direction, which resulted in the motion oscillating until it stopped. This again is due to the dampening causing the forces to slowly reach zero. The most significant observation from this is that the dampening force and the gravity (the tilt that our accelerometer was at) did not drive the motion and the Force of the spring drove the motion of the ball in contrast to experiment 3 where the gravity drove the motion. This is

due the system being overdamped. The figure below shows these forces together, and figure 8 shows the motion that the ball took when we did this experiment.

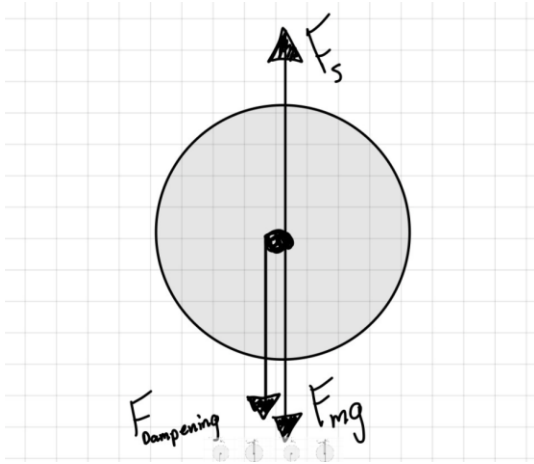


Figure 7. Free Body Diagram of The three forces, with dampening and gravity forces opposing the spring force in Exercise 4

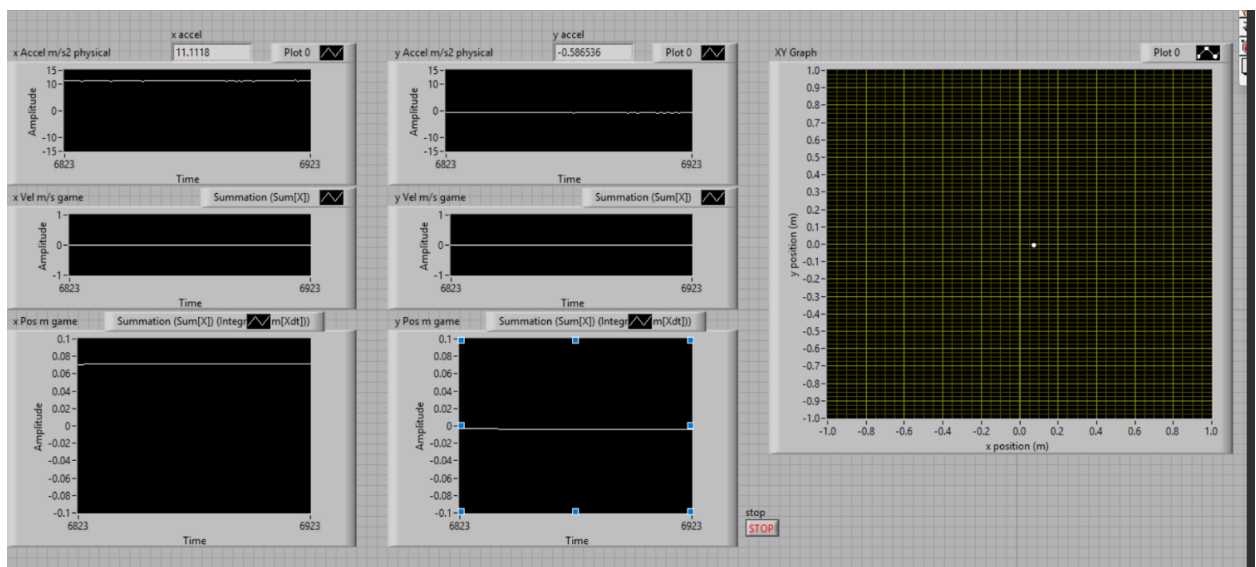


Figure 8. graph of the ball's motion from the front panel of exercise 4

3.8 Exercise 5

For this exercise the tilting was the same as previous procedures, however the ball was moving in a circle from just tilting the accelerometer. This was due to the acceleration of the spring force being equal to the acceleration of the virtual ball. This can be seen in the free body diagram below, with the tangential velocity of the ball and the force of the spring.

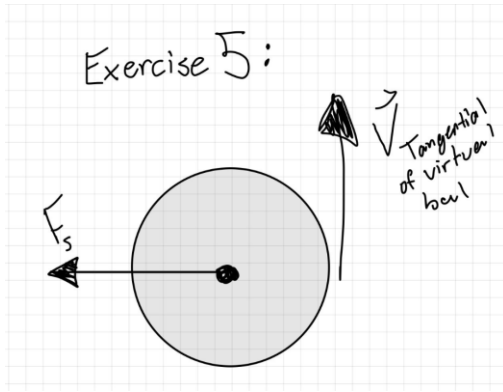


Figure 9. Free body diagram of the force of the spring acting on the mass

3.9 Exercise 6

Finishing with the last exercise, the dampening coefficient was changes to 0.2, which led to the virtual ball moving in a perfect circle. Similar to exercise 5, however the ball would spiral back to the center and this was du to the dampening, because the acceleration of the ball was slowly progressing from the dampening, leading to the spring force being greater than the radial acceleration, as show from the free body diagram.

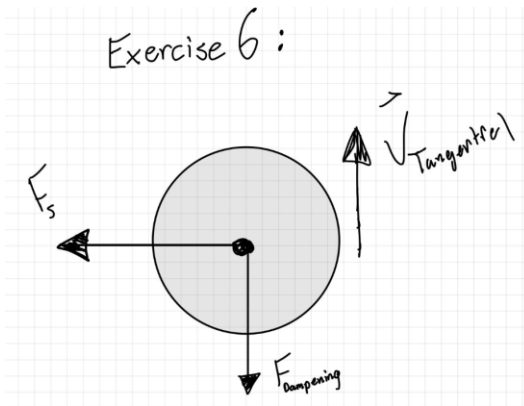


Figure 10. Free body diagram of forces acting on virtual Mass

4.0 Conclusions

The bias levels for this lab included 1.645V and 1.654V for the accelerometer at 3.3V in the x and y ais. These values were not with the given range from the data sheet for the accelerometer at 3V, but this behavior is expected as the accelerometer experiences ratiometric behavior, which means since the accelerometer has more voltage the other values, such as the bias voltage will also increase. The sensitivities measures were denoted as 315 mV/g and

342mV/g in the x and y axis, which again were ranged pretty high in comparison to the given range from the data sheet. Lastly, we can say our measurements were consistent and reliable as the sampling frequency was twice the value of the accelerometer's bandwidth which was 100Hz and 50Hz, which means the Nyquist criterion was not violated.